

Tokenization: Different ~~and~~ LLMs have different tokenizer approaches

~~space~~ how many spaces are considered as
~~numeric~~ how many numeric characters & tokens
letters are considered as a token

Compression ratio: 1 token = ~~1.6~~ byte
1.6

Byte-tokenizer / character-tokenizer
① word-tokenizer
character Byte-pair encode - tokenizer

↓
GPT 2, 3 ... etc --

Start with each byte as a token,
and then merge the most common
pair of adjacent tokens

Lecture #2: Build LLM with PyTorch.

#1: All primitives to train a model

#2: bottom-up ~~from~~ [tensors to model to optimizer]

#3: efficiency (use of resources) $\rightarrow$ training loop

$\Downarrow$

(15 TB)
Num of Tokens

(70B) Model size $\Leftrightarrow$ compute $\xrightarrow{1024}$ ~~1000~~ H100s

example:

total flops: $6 * 70^{e9} * 15^{e12} \downarrow$ num of tokens
 $\downarrow$ num of param.

H100_flop_per_sec = $1979^{e12}/2$.

$\downarrow$
mfu = 0.5

$\downarrow$
Flops_per_day = H100_flop_per_sec * mfu * $\overset{\text{Num of GPU}}{1024} * \overset{\text{days for secs}}{60 * 60 * 24}$

$\downarrow$
days = total_flops / flops_per_day

Q: how many hours to train a model?

Q: what's the largest model that you can train on ...?